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ON DESIGN CONSIDERATIONS AND RANDOMIZATION-BASED INFERENCE FOR COMMUNITY INTERVENTION TRIALS

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SUMMARY

This paper discusses design considerations and the role of randomization-based inference in randomized community intervention trials. We stress that longitudinal follow-up of cohorts within communities often yields useful information on the effects of intervention on individuals, whereas cross-sectional surveys can usefully assess the impact of intervention on group indices of health. We also discuss briefly special design considerations, such as sampling cohorts from targeted subpopulations (for example, heavy smokers), matching the communities, calculating sample size, and other practical issues. We present randomization tests for matched and unmatched cohort designs. As is well known, these tests necessarily have proper size under the strong null hypothesis that treatment has no effect on any community response. It is less well known, however, that the size of randomization tests can exceed nominal levels under the 'weak' null hypothesis that intervention does not affect the average community response. Because this weak null hypothesis is of interest in community intervention trials, we study the size of randomization tests by simulation under conditions in which the weak null hypothesis holds but the strong null hypothesis does not. In unmatched studies, size may exceed nominal levels under the weak null hypothesis if there are more intervention than control communities and if the variance among community responses is larger among control communities than among intervention communities; size may also exceed nominal levels if there are more control than intervention communities and if the variance among community responses is larger among intervention communities. Otherwise, size is likely near nominal levels. To avoid such problems, we recommend use of the same numbers of control and intervention communities in unmatched designs. Pair-matched designs usually have size near nominal levels, even under the weak null hypothesis. We have identified some extreme cases, unlikely to arise in practice, in which even the size of pair-matched studies can exceed nominal levels. These simulations, however, tend to confirm the robustness of randomization tests for matched and unmatched community intervention trials, particularly if the latter designs have equal numbers of intervention and control communities. We also describe adaptations of randomization tests to allow for covariate adjustment, missing data, and application to cross-sectional surveys. We show that covariate

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adjustment can increase power, but such power gains diminish as the random component of variation among communities increases, which corresponds to increasing intraclass correlation of responses within communities. We briefly relate our results to model-based methods of inference for community intervention trials that include hierarchical models such as an analysis of variance model with random community effects and fixed intervention effects. Although we have tailored this paper to the design of community intervention trials, many of the ideas apply to other experiments in which one allocates groups or clusters of subjects at random to intervention or control treatments.

1. INTRODUCTION

This paper concerns community intervention trials and other trials in which groups of subjects are allocated at random to treatment. Randomization of subjects in groups, rather than of each individual separately, has important consequences for interpretation and analysis.[1] The groups of subjects are likely to be heterogeneous, giving rise to a component of variation that one must take into account in the analysis and that one can control only by studying many groups, rather than many subjects within each group. In this paper we refer to communities as the group allocated at random to treatments, but the same principles apply to other types of groups, such as neighbourhoods, factories, clinical practices, schools or classrooms.

Surveys in which one selects at random clusters of subjects resemble community intervention trials in that one must take into account the heterogeneity among clusters.[1] For example, there have been surveys conducted to evaluate school performance by gathering information on the pupils who enter, school characteristics, and academic performance.[2] In such studies, randomly selected schools represent the 'cluster' or 'group'. Such surveys, however, are usually observational rather than experimental, and in this respect they differ from community intervention trials, in which the allocation of treatment to community is under experimental control. For example, the school characteristics examined, such as school size, are not usually under experimental control. 'Ecologic' studies, in which one analyses health responses of groups of people in given geographic areas as if everyone in an area were subject to a common exposure (for example, average fat consumption), are also observational. There may be confounding of apparent exposure effects in such studies with other unmeasured exposures in these geographic areas. Thus community intervention trials are related to, but in important respects are different from, certain observational study designs.

Many papers have appeared on the design and analysis of community intervention trials. These papers have dealt with approaches to hypothesis testing,[1,3–12] estimation,[13,14] covariate adjustment,[10,15–19] sample size,[10,20–25] the usefulness of matching the communities,[10,26,27] and other issues in design.[9,10,28] Moreover, there is an important related literature that concerns models used to describe the group variation as well as the effects of individual level covariates.[2,29–31] Most of these papers have utilized model-based approaches to inference, for example, by introducing a random community effect in an analysis of variance. Maritz and Jarrett[6] and Gail *et al.*,[10] however, emphasize randomization-based inference.

One purpose of the present article is to clarify possible objectives of group randomized trials and to relate these objectives to available options in design and analysis, such as whether to use cohorts or cross-sectional surveys. These objectives and some implications for choosing cohort or cross-sectional designs appear in Section 2. A second purpose is to present and study methods for randomization-based inference, including permutation tests, for cohort designs (Section 3) and cross-sectional designs (Section 5). In particular for cohort designs, we review underlying assumptions, distinguish between the 'strong' null hypothesis of no treatment effect on any community response and the 'weak' null hypothesis of no treatment effect on average.[32] Fisher[33]

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and most texts on randomization or permutation tests[34-36] focus on the 'strong' null hypothesis. The weak null hypothesis,[32] however, is relevant to community intervention trials because one would not ordinarily want to adopt a treatment that produced no improvement in average community response, even though it might influence some communities favourably and others unfavourably, in violation of the strong null hypothesis. We study the size of randomization tests under the weak null hypothesis both for unmatched (Section 3.1) and for matched designs (Section 3.2). We study the power of randomization tests with covariate adjustment (Section 3.3), the impact of missing data on randomization-based inference (Section 4), and randomization-based inference for cross-sectional studies (Section 5). We have not found previous simulation studies on the size of randomization tests under the weak null hypothesis that are tailored to community intervention trials, nor have we seen previous studies of the power of randomization tests for community intervention trials with covariate adjustment. Thus, much of the material in Sections 3, 4 and 5 is not available in the literature. Since the work of Neyman *et al.*,[32] however, it has been appreciated that randomization tests (also called permutation tests) need not have valid size under the weak null hypothesis, and Boik[37] and Romano[38] showed that the size of the permutation test for analysis of variance can exceed nominal levels if variances are heterogeneous.

The reader interested in general aspects of design and on the practical implications of our simulation studies for randomization-based analysis of community intervention trials may wish to read Sections 2, 4, 5 and 6. The reader interested in additional information on the statistical properties and robustness of randomization-based inference and on covariate adjustment may wish to study Section 3.

2. SOME DESIGN CONSIDERATIONS

2.1. Rationales for Using Group Randomization

Because individual randomization usually yields more information per subject studied than group randomization,[1] one might wonder why group randomized designs are used. We present two examples to illustrate various rationales for selecting a group randomized design.

Sommer, *et al.*[39] randomly allocated 229 villages in Sumatra to intervention with vitamin A supplements and 221 villages to serve as controls. Preschool children over age 1 were identified and examined at baseline, and eligible children were given vitamin A by mouth at about 2 and 7 months after baseline examination in the intervention community. Eligible preschool children were similarly identified and examined at baseline in the control villages. The vital status of the children in these 450 village cohorts was assessed at 11–13 months. Vitamin A supplementation was found to reduce total mortality rates by 34 per cent from 7·3 to 4·9 per 1000 person-years. 'Political and administrative reasons' were cited as the rationale for this design. It was not acceptable to treat some children in a given village and not others. Thus, even though, in principle, an individually randomized trial that compared vitamin A supplementation with placebo was conceivable, it was not feasible in this setting.

Another aspect of feasibility is cost. One can imagine, for example, a complicated treatment that requires the availability of an operating room in case of an emergency. One might reduce costs substantially by treating groups of patients, because elaborate medical facilities need only be located in the intervention communities.

Quite apart from political feasibility or cost, there may be no alternative to group randomization because the intervention will necessarily affect all members of a group. For example, the Community Intervention Trial for Smoking Cessation (COMMIT) was designed to study interventions to accelerate smoking cessation among heavy smokers[10,19] and to reduce smoking

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prevalence.[10,40] Health care providers and employers were encouraged to promote smoking cessation, and public education and media campaigns were undertaken. These interventions were community-based and had the potential to affect every smoker in the community. Thus, the intervention precluded individual randomization within communities. Instead, communities were matched into eleven pairs, with one community in each pair randomly allocated to active intervention.

Thus, despite some loss of statistical efficiency per subject studied, it may be desirable, or necessary, to choose a group randomized design, either: (1) for purposes of political feasibility; (2) to reduce costs; or (3) because intervention will necessarily affect all members of a group.

2.2. Study Objectives and the Choice Between Longitudinal Cohort Studies or Serial Cross-Section Surveys

Usually it is not feasible to study every member of a community. Instead, one gauges the effects of treatment either by studying a selected cohort drawn from within each community and obtaining longitudinal measurements on cohort members or by measuring health status through serial cross-sectional surveys. For helpful comparisons of longitudinal and cross-sectional designs, see Duncan and Kalton[41] and Dwyer and Feinleib.[42]

Suppose one desires to compare changes from baseline in treated and untreated communities. At time t_0 (baseline) one might randomly select a cohort from within each community and measure the health criterion, such as blood pressure or smoking status at baseline, Y_{ijkt_0}. The quantity Y_{ijkt_0} corresponds to the kth member in the cohort selected in the jth community assigned to treatment $i(i = 1, 2; j = 1, 2, \ldots, J; k = 1, 2, \ldots, K_{ij})$. At the end of the experiment (t_1), one obtains a final measurement Y_{ijkt_1} on each cohort member. One could use $Y_{ij\cdot t_1}$, the average level at the end of the experiment or $Y_{ij\cdot t_1} - Y_{ij\cdot t_0}$, the average change in Y among cohort members in community (i, j) to summarize data from the experiment in that community. (Throughout this paper, a subscripted 'dot' indicates a simple average over the corresponding index.)

One important option associated with cohort studies is that one need not choose the cohorts to be representative of entire study communities. Instead, one can select the cohorts to measure the targeted effects of treatment. For example, in the COMMIT study, cohorts of approximately 450 randomly selected heavy smokers were recruited and followed in each community to determine whether intervention could increase the proportion of heavy smokers who quit. These cohorts were not a random sample from the entire community, but were a targeted random sample of heavy smokers in each community. Letting $Y_{ijkt_1} = 1$ if the smoker had quit and zero otherwise, and noting that all cohort members were smokers at baseline $(Y_{ijkt_0} = 0)$, the change $Y_{ijkt_1} - Y_{ijkt_0} = Y_{ijkt_1}$ indicates whether the smoker quit, and $Y_{ij\cdot t_1} - Y_{ij\cdot t_0} = Y_{ij\cdot t_1}$ is the cohort quit rate in community (i, j).

Consider instead a cross-sectional design in which $Y^*_{ijkt_0}$ corresponds to the kth value in a random sample of size K_{ij} drawn from community (i, j) at baseline and $Y^*_{ijkt_1}$ is a value in an *independent* random sample from the population in the community at t_1. We can use the averages $Y^*_{ij\cdot t_1}$ or $Y^*_{ij\cdot t_1} - Y^*_{ij\cdot t_0}$ to characterize health effects in community (i, j). In the COMMIT example, $Y^*_{ij\cdot t_1}$ corresponds to the prevalence of heavy smoking in community (i, j) after intervention, and $Y^*_{ij\cdot t_1} - Y^*_{ij\cdot t_0}$ corresponds to the decrease in smoking prevalence.

Suppose that the initial population in community (i, j) at t_0 was protected from dying and not allowed to leave the community and that no new people were allowed to enter the community, either by immigration or by birth. Suppose further that the cohort was drawn as a representative random sample from the community at t_0, rather than as a targeted cohort. Then Y_{ijkt_0} and

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$Y^*_{ijkt_0}$ are each a random observation from the 'community' at time t_0, and Y_{ijkt_1} and $Y^*_{ijkt_1}$ are each a random observation at t_1 from the population first identified at t_0. Thus, Y_{ijkt} and Y^*_{ijkt} estimate the same quantities at t_0 and t_1.

In general, however, Y_{ijkt} and Y^*_{ijkt} do not estimate the same quantities. Some members of the original community at time t_0 die before t_1, others are born into the community, others emigrate, while still others immigrate. Thus the community sampled at t_1 in the cross-sectional survey may differ somewhat from the surviving members at t_1 of the original cohort sampled at t_0. The surviving cohort members are necessarily older by an amount $t_1 - t_0$, whereas the average age of community members may not have changed. Many cohort members may have been exposed to the intervention for the entire study period, whereas subjects who immigrate into the community shortly before t_1 will have had little chance for exposure to intervention. The differences between the cohort and cross-sectional survey data are still greater if the cohort was 'targeted', rather than chosen completely at random from the community at t_0. Thus, estimates of treatment effects from cohort or cross-sectional studies need not agree, and it is important to understand the scientific objectives of the experiment so as to choose properly between the cohort and cross-sectional designs.

Consider the following three possible objectives:

1. assess the effect of an intervention directed at specific individuals on those individuals;
2. assess the effect of a community-based intervention on individuals;
3. assess the effect of a community-based intervention on group indices of health.

Studies of cohorts at which intervention is directed are best suited for objective 1, studies of targeted cohorts or studies of cohorts representative of the entire community are best suited to objective 2, and repeated cross-sectional samples or studies of representative cohorts are suitable for objective 3.

In many studies, the primary objective is to determine the effects of an intervention directed at specific individuals on those individuals (objective 1). For example, in the vitamin A study, the principal objective is to determine whether vitamin A supplementation lowers a Sumatran child's chance to death (and blindness). Determination of whether vitamin A supplementation lowers community death rates, while of public health importance, seems secondary to determination of the effect of vitamin A at the individual level. The cohort design used in the vitamin A study was well-suited to study objective 1, because: ineligible children (with severe disease) were excluded; baseline information was gathered on individuals; treatment was directed specifically at cohort members, and there was assessment of compliance to treatment; and follow-up was attempted to determine the vital status at 1 year for each subject in the cohort.

Likewise, a principal objective in the COMMIT study is to determine whether community-based interventions, even though not exclusively directed at heavy smokers, can accelerate the tendency to quit smoking among heavy smokers. From this perspective, objective 2 seems more important than objective 3, and the cohort design seems apposite. However, the cohorts of heavy smokers were selected as representative of all heavy smokers in each community, there was study of additional representative cohorts of light and moderate smokers, and there were precautions taken to assure that no special intervention efforts were directed at cohort members, whose identities were not revealed during the study. Thus these cohorts also provide information on the effectiveness of the intervention at the community level (objective 3).

In addition, COMMIT relied on independent baseline and final cross-sectional surveys to give an overall community-level assessment of progress in reducing the prevalence of smoking (objective 3). Such changes in prevalence reflect, in part, the effects of the intervention programme

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on individuals, such as whether intervention induces smokers to quit or non-smokers to remain non-smokers. Changes in smoking prevalence, however, may also reflect patterns of immigration or emigration that are correlated with the effects of intervention at the individual level. As an extreme example, suppose an intervention offers smokers early retirement and an ample pension, provided that they move elsewhere. Such an intervention could lower smoking prevalence (objective 3) without causing any individual to stop smoking.

Objectives 2 and 3 may be more meaningful in settings where the response of one individual within a community affects the response of another individual. For example, suppose that one plans to test a new vaccine (vaccine A) to prevent a highly infectious disease in intervention communities. Subjects in control communities will receive a standard vaccine B. Suppose it has been demonstrated in individually randomized trials that vaccine A is more effective than vaccine B in preventing infection following an infectious challenge, but suppose vaccine A has side-effects such as malaise that cause many individuals to refuse vaccination, whereas vaccine B has few side-effects and a high acceptance rate. In this setting, control communities may have lower infection rates, both among unvaccinated and vaccinated members, than intervention communities, because a higher proportion of members of the control communities will become vaccinated, inducing greater herd immunity. Serial cross-sectional surveys of disease prevalence or follow-up of representation intervention cohorts that include some unvaccinated members provide reasonable group indices of the overall effectiveness of the vaccination campaign (objective 3), and studies of representative cohorts can also provide information on the effectiveness of this community-based intervention at the individual level (objective 2).

Occasionally, cohort studies cannot provide data required for objective 3. For example, suppose one wishes to determine the effect of a 20 year community-based intervention on the prevalence of smoking among 15 year olds. A cohort assembled at the beginning of the study provides no information on this age group at the end of the intervention. The only means to obtain information on changes in this index of group health is with cross-sectional surveys at the beginning and end of the study.

Another consideration that affects the choice between longitudinal and cross-sectional designs is statistical efficiency. Suppose, for example, that a measure of the effect of intervention in a longitudinal study is $(Y_{1\cdot\cdot t_1} - Y_{1\cdot\cdot t_0}) - (Y_{2\cdot\cdot t_1} - Y_{2\cdot\cdot t_0})$. Because Y_{ijkt_1} and Y_{ijkt_0} tend to be positively correlated, this measure of treatment effect usually has smaller variance than the corresponding quantity estimated from two independent cross-sectional surveys, $(Y^*_{1\cdot\cdot t_1} - Y^*_{1\cdot\cdot t_0}) - (Y^*_{2\cdot\cdot t_1} - Y^*_{2\cdot\cdot t_0})$, in which $Y^*_{ijkt_1}$ and $Y^*_{ijkt_0}$ are uncorrelated.[9,11] Efficiency gains can be dramatic[10] when the cohort chosen for longitudinal study is targeted to a subset of the population, such as heavy smokers. Thus, longitudinal studies tend to have higher statistical efficiency per subject studied than do cross-sectional designs.

To summarize, studies of cohorts given specific intervention or control treatments yield individual-level data on baseline covariates, treatment compliance and outcome, and are well-suited to determine treatment effects at the individual level (objective 1). Studies of targeted cohorts or representative cohorts are well-suited to determine the effects of a community-based intervention on individuals (objective 2). Serial cross-sectional surveys or studies of representative cohorts are adequate to gauge the overall public health impact of an intervention (objective 3). In selecting a design, however, one must also consider statistical efficiency and the feasibility of overcoming potential practical problems, such as loss to follow-up in cohort studies and non-response in cross-sectional surveys. Discussion of the important topic of loss to follow-up in cohort studies and possible resulting biases appears in Sections 2.3 and 4, and corresponding material on non-response bias in cross-sectional surveys appears in Sections 2.3 and 5.

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2.3. Matching, Sample Size and Other Design Considerations

In balanced, unmatched designs, one randomly selects J of $2J$ communities as intervention communities and J to serve as control communities. There are $(2J)!/(J!)^2$ equally likely allocations. In assessing the results of such a trial, one must consider two sources of random variation: (1) intrinsic heterogeneity among communities, and (2) sampling variation that arises when one selects a cohort or repeated cross-sectional samples from within a community. If the variability associated with intercommunity heterogeneity is large, the best way to increase the power of an unmatched experiment is to increase the number of communities studied. Additionally, one can attempt covariate adjustment or post-stratification on community characteristics.

Another way to control intercommunity heterogeneity is to group communities into nearly homogeneous strata (or 'blocks') and to allocate equal numbers of communities at random to intervention or control arms within each block. The most popular version of this strategy is to match communities in pairs (blocks of size two). Then there are J blocks and 2^J equally likely treatment allocations of the $2J$ communities. Whether matching leads to a gain or loss of power depends on how large the intercommunity heterogeneity is, compared to the within community sampling variation, and also on how effectively the matching procedure reduces community heterogeneity within community pairs.[10, 26, 27]

One of the more problematic aspects of designing community intervention trials is the lack of data on intercommunity heterogeneity.[43] Sometimes, as in COMMIT, one can rely on related experiments to gain some idea of this component of variation,[10] but, because this component of variation is uncertain and often larger than the within community sampling variation, it is necessary in selection of a sample size to explore a range of possible values of intercommunity heterogeneity.

Likewise, when designing a community intervention trial one may have little quantitative information on the effectiveness of matching. In this case, a conservative strategy is to design the matched study on the assumption that matching is completely ineffective, namely completely at random. This was the strategy employed in designing COMMIT.[10]

The following design considerations, while not solely relevant to community intervention trials, are essential to the success of such trials. One should make every effort to limit loss to follow-up in cohort studies. Useful tactics include obtaining initial informed consent for follow-up, using special procedures to attempt to find lost subjects, and limiting questionnaires to essential items. It may be useful to gather covariate information and intercurrent responses to permit statistical adjustments for losses to follow-up (Section 4). One should make every effort to treat members of a cohort no differently from other members of the community from which they had been selected.

Non-response to a solicitation for information in a cross-sectional design can induce biases analogous to those that arise from loss to follow-up in a cohort study, and efforts to minimize such non-response are essential. Furthermore, it may be difficult to establish a sampling mechanism that yields a representative sample of the entire community.

It is vital that the endpoint chosen is clearly defined, easily determined, and convincing to the scientific community. For example, in COMMIT, the definition of smoking cessation was a self-report of no smoking during the six months preceding the end of the study.[19] In large measure, the credibility of this study depends on the acceptability of this endpoint. Although there were some biochemical studies undertaken to determine whether self-described quitters in intervention communities tended to misrepresent their smoking status at the end of the study more than did self-described quitters in control communities, it was not feasible to monitor smoking status biochemically over extended periods. Therefore the main endpoint of this study remains self-reported quit rates.

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Particularly in cohort studies, it may be useful to gather individual level and community level covariate information at baseline. One can use such data to increase the power of randomization-based hypothesis tests (Section 3.3) and of model-based analyses that also estimate the effects of covariates on response.

3. ANALYSES FOR COHORT DESIGNS

In Sections 3.1 and 3.2 we define randomization tests for cohort designs and present simulation studies on their size under the 'weak' null hypothesis of no treatment effect on average. We review the practical implications of these simulations in the Discussion (Section 6). Section 3.3 describes methods for randomization tests with covariate adjustment and related simulations on gains in power from covariate adjustment. Again, we review practical conclusions in Section 6. Readers interested only in practical implications for design and analysis may wish to skip Section 3.

3.1. Randomization-Based Inference for Unmatched Designs

As in Section 2.2, let Y_{ijkt} denote the response of cohort member k in community j on treatment i at time t for $i = 1, 2$, $j = 1, 2, \ldots, J$, $k = 1, 2, \ldots, K_{ij}$, and $t = t_0, t_1, \ldots, t_T$. Unless otherwise noted, we assume equal numbers of communities are assigned to intervention and control groups. Here t_0 corresponds to baseline and t_T to the end of the experiment. Let W_{ijk} be a scalar function of Y_{ijkt} for $t = t_0, t_1, \ldots, t_T$ that summarizes the health effects on cohort member (i, j, k). For example, $W_{ijk} = Y_{ijkt_T}$ is the final health response, and $W_{ijk} = Y_{ijkt_T} - Y_{ijkt_0}$ measures the change from baseline. We could also use the least squares estimate of the slope of the sequence of measurements at times $t_0, t_1, \ldots, t_T$ for W_{ijk}. The quantity W_{ijk} can be continuous or discrete. In the COMMIT example of Section 2, $W_{ijk} = Y_{ijkt_1} - Y_{ijkt_0} = Y_{ijkt_1}$ was a dichotomous indicator of whether a person who was smoking at baseline ($Y_{ijkt_0} = 0$) had quit smoking by the end of the experiment t_1.

Unless otherwise noted, we assume that available measurements are representative, namely that missingness is completely at random[44] and that there are K_{ij} individuals in community (i, j) with sufficient information to compute W_{ijk}. Thus $W_{ij.} \equiv K_{ij}^{-1}\sum_k W_{ijk}$ is an unbiased estimate of the expected cohort response.

3.1.2. Testing the 'Strong' Null Hypothesis

We begin by reviewing Fisher's classic randomization-based approach[33] to hypothesis testing. Consider the statistic

$$U \equiv J^{-1}\sum_j (W_{1j.} - W_{2j.}) \equiv W_{1..} - W_{2..}. \tag{1}$$

We can use a randomization test[33] to test the 'strong null hypothesis', H_{0s}: treatment has no effect on any of the community means $W_{ij.}$. Under this hypothesis, and conditional on the observed values $\{W_{ij.}\}$, all reallocations of communities to treatments are equally likely. Thus, we can generate the randomization (permutational) distribution of U by reallocating the observed $\{W_{ij.}\}$ to treatment in $(2J)!/(J!)^2$ ways and computing U for each such reallocation. If the originally observed U is greater than the $1 - \alpha$ quantile of the permutational distribution, then we reject the null hypothesis H_{0s} at the significance level α in favour of the one-sided alternative that treatment $i = 1$ (intervention) leads to larger values of W than does treatment $i = 2$ (control). We can devise two-sided tests by using the $\alpha/2$ and $1 - \alpha/2$ quantiles as critical values of U.

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Because the permutational distribution is discrete, there may be no point of increase that corresponds exactly to the desired significance level, α. In this case, one chooses the critical value to assure that the size of the test is less than α. For example, with $J = 4$ the permutational distribution function has 70 points of increase, assuming no ties. Rejection with a one-sided test at the 0·05 level occurs if U is among the three largest values of the permutational distribution, because $3/70 = 0{\cdot}043$ but $4/70 = 0{\cdot}057$. The issue of discreteness becomes unimportant as the number of communities increases.

One can also base interval estimates of treatment effect on randomization tests, as for example in Section 1.4 of the text by Manly.[36] Assume that the effect of intervention ($i = 1$) is to increase the average response by δ, compared to control ($i = 2$), for every community.[45] In particular, there is no treatment–community interaction. Thus, if one of the communities labelled $(2, j)$ had, instead, received treatment $i = 1$, the response would be $\delta + W_{2j.}$. Likewise, a community labelled $(1, j)$ would yield a response $W_{1j.} - \delta$ if, instead, it had received treatment $i = 2$. One can construct a test-based confidence interval on δ by considering the quantities $W^*_{1j.} = W_{1j.} - \delta$ and the corresponding statistic $U^* = W^*_{1..} - W_{2..} = U - \delta$. One obtains the permutational distribution of U^* from reallocation of the $\{W^*_{1j.}\}$ and $\{W_{2j.}\}$ to treatment. The upper limit of the confidence interval for δ is δ_u, the smallest value of δ such that U^* is less than the $\alpha/2$ quantile of its permutational distribution, and δ_l, the lower limit of the confidence interval, is the largest value of δ such that U^* exceeds the $1 - \alpha/2$ quantile of its permutational distribution. The resulting confidence interval (δ_l, δ_u) has level α. This procedure was used to place a confidence interval on the difference in smoking quit rates between intervention and control communities in the COMMIT trial.[19]

With more than 30 communities, one will find it very time-consuming or even impossible to compute the randomization distribution of U exactly, because $30!/(15!)^2$ exceeds $1{\cdot}55 \times 10^8$. However, the unpaired t-statistic,

$$t = U \left\{ \frac{(J_1 - 1)s_1^2 + (J_2 - 1)s_2^2}{(J_1 + J_2 - 2)} \left(\frac{1}{J_1} + \frac{1}{J_2} \right) \right\}^{-1/2} \tag{2}$$

on $J_1 + J_2 - 2$ degrees of freedom provides an excellent approximation[46] to the permutational distribution of U under H_{0s}. Here J_1 and J_2 are the numbers of communities assigned to intervention $i = 1$ or 2, respectively, $U = W_{1..} - W_{2..}$, $s_1^2 = (J_1 - 1)^{-1} \sum_j (W_{1j.} - W_{1..})^2$ and s_2^2 is similarly defined. Note that if $J_1 = J_2 = J$, equation (2) reduces to $U\{(s_1^2 + s_2^2)/J\}^{-1/2}$, and, as J increases, $(s_1^2 + s_2^2)/J$ converges under H_{0s} to $2\sigma_p^2/J$, where $\sigma_p^2 \equiv \sum_{i,j} (W_{ij.} - W_{...})^2/2J$ is the variance of the finite population of community responses, $\{W_{ij.}\}$. For large J_1 and J_2, the permutational distribution of U is approximately normal[47] with mean 0 and variance $\omega = \sigma_p^2 (J_1 + J_2)^2 / J_1 J_2 (J_1 + J_2 - 1)$. We define the 'normal approximation deviate' as $z = U\omega^{-1/2}$, and the 'normal approximation test' (Table I) rejects when z exceeds 1·645.

3.1.3. *Validity of Testing the Null Hypothesis of No Intervention Effect on Average*

As Neyman *et al.*[32] observed, we may not have interest in the strong null hypothesis of no treatment effect on any community mean, H_{0s}, but rather in the null hypothesis H_0 that the *average* treatment effect is zero. For example, suppose

$$W_{ijk} = \mu + d_{ij} + \varepsilon_{ijk} \tag{3}$$

where $\varepsilon_{ijk} \sim N(0, \sigma^2)$ is a normal variate with mean 0 and variance σ^2 that represents individual variation within community. Also $d_{ij} \sim N(\delta_i, \phi_i)$ represents the effect of treatment i in community

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Table I. Estimated size times (1000) of the one-sided permutation test (upper number), unpaired t-test (middle number) and normal approximation test for various variance ratios $\psi = \mathrm{var}(W_{2j.})/\mathrm{var}(W_{1j.})$†

Number of communities		ψ						
Intervention J_1	Control J_2	1/5	1/3	1/2	1	2	3	5
3	3	61	41	46	46	56	46	67*
		54	41	55	48	44	44	64
		53	40	53	44	44	43	60
5	5	52	61	52	53	47	47	46
		54	62	58	51	48	51	53
		54	62	58	52	48	51	53
10	10	50	55	51	49	54	50	51
		47	50	55	47	52	45	45
		48	51	56	47	53	45	45
3	6	95*	67*	55	48	32	28	44
		106*	76*	57	54	37	28	36
		106*	76*	57	54	37	28	36
6	3	24	28	41	43	74*	65*	85*
		19	27	39	47	81*	74*	100*
		19	27	39	47	81*	74*	100*
60	30	18	27	42	44	68*	82*	105*
		17	27	42	44	68*	81*	108*
		17	27	42	44	68*	81*	108*

* An asterisk indicates that the observed size exceeds the nominal significance level by 20 per cent or more and is outside the interval (36, 64) which should include 95 per cent of the observations if the size is 0·05

† Estimates are based on 1000 independent simuated replicates for each combination of (J_1, J_2) and Ψ. Estimates outside the interval (36, 64) differ significantly from the nominal 0·05 level, based on a two-sides 0·05 level test. Each value of (J_1, J_2, Ψ) defines an independent simulation study. Calculations were performed in Fortran 77 (WATCOM compiler) on a personal computer with a Pentium™ microprocessor. Normal pseudo-random numbers were obtained from the Box–Muller algorithm described as the 'direct method' on page 953 in the book edited by Abramowitz and Stegun.[57] Uniform pseudo-random numbers needed for this algorithm were obtained as in the footnote to Table II

j. We let $\delta_2 = 0$, and assume $\{d_{ij}\}$ and $\{\varepsilon_{ijk}\}$ are mutually independent. In this model, Neyman's null hypothesis H_0 is $\delta_1 = 0$. Note that under this null hypothesis, $\mathrm{var}(W_{1j.}) = \phi_1 + \sigma^2/K_{1j}$ need not equal $\mathrm{var}(W_{2j.}) = \phi_2 + \sigma^2/K_{2j}$, even if $K_{1j} = K_{2j}$. For finite samples, the permutation test can have a size that exceeds the nominal α level under H_0, making it inappropriate. Under the balanced design with J communities on each treatment group, however, the unpaired t-test will have the correct size as the number of communities increases, because s_1^2 and s_2^2 converge to $\mathrm{var}(J^{1/2} W_{1..}) = \phi_1 + J^{-1} \Sigma\, \sigma^2/K_{1j}$ and $\mathrm{var}(J^{\frac{1}{2}} W_{2..}) = \phi_2 + J^{-1} \Sigma\, \sigma^2/K_{2j}$, respectively, and the t-statistic is therefore properly standardized for convergence to a standard normal deviate (see Miller,[48] Section 2.3). The normal approximation to the randomization test above also yields a valid test of H_0 for large J, as does the permutation test itself. This nice property fails to hold if one assigns different numbers of communities to treatments $i = 1$ or 2, however, both for the unpaired t-test, for the normal approximation and for the randomization test.

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A simple example illustrates that the permutation test need not have nominal size under H_0 in small samples. Assume that $E(W_{1j.}) = E(W_{2j.}) = 0$, $\text{var}(W_{1j.}) = 10,000$ and $\text{var}(W_{2j.}) = 0{\cdot}01$ and that there are only four communities. Suppose that the outcomes for communities $(i, j) = (1, 1), (1, 2), (2, 1), (2, 2)$ are, respectively, 100, 200, 0·2, $-0{\cdot}1$. Thus $JU = 2U = 299{\cdot}9$. The permutational distribution of $2U$ takes on the six values $-299{\cdot}9$, $-100{\cdot}3$, $-99{\cdot}7$, $99{\cdot}7$, $100{\cdot}3$, $299{\cdot}9$. Thus, we would reject H_{0s} at the one-sided $\alpha = 1/6$ level. Because $E(W_{1j.}) = E(W_{2j.})$, the null hypothesis H_0 is true in this example. The strong null hypothesis H_{0s}, however, is not true because treatment affects the variance of $\{W_{ij.}\}$. Rejection of H_{0s} occurs only if the two largest $W_{ij.}$ values occur on treatment $i = 1$. Under H_0, rather than H_{0s}, two large positive values on treatment $i = 1$ are likely to occur about one-fourth of the time, leading to an actual size of the randomization test of about 1/4. To be precise, both $|W_{11\cdot}|$ and $|W_{12\cdot}|$ will greatly exceed $|W_{21\cdot}|$ and $|W_{22\cdot}|$ with high probability, and the chance that $W_{11\cdot}$ and $W_{12\cdot}$ are both positive is $\frac{1}{4}$.

To determine how misleading the randomization test can be for testing H_0 in more realistic cases, we simulated from the model (3), using the parameters $\mu = \delta_1 = \delta_2 = 0$, $\text{var}(W_{1j.}) = 1$ and $\text{var}(W_{2j\cdot}) = \psi$, for $\psi = \frac{1}{5}, \frac{1}{3}, \frac{1}{2}, 1, 2, 3, 5$. We estimated the size of the permutation test, the normal approximation test and the unpaired t-test from 1000 independent replications using nominal 0·05 level one-sided rejection regions. We conducted independent sets of simulations for each different set of values of J_1, J_2 and ψ. We evaluated the randomization distribution exhaustively for all cases in Table I except the case with $J_1 = 60$ and $J_2 = 30$. In this case we estimated the distribution from a random sample of 100,000 independent reallocations of $W_{ij\cdot}$ to treatments.

These simulations indicate that the size of permutation tests is near nominal levels for balanced designs ($J_1 = J_2$) for values of ψ likely to arise in practice, even with only three communities in each treatment group. For unbalanced designs, the one-sided permutation test has a size that exceeds the nominal 0·05 level if $J_1 < J_2$ and $\psi < 1$ or if $J_1 > J_2$ and $\psi > 1$. For example, with $J_1 = 6$ and $J_2 = 3$, the estimated size is 0·065 for $\psi = 3$ and 0·085 for $\psi = 5$. With $J_1 = 60, J_2 = 30$ and $\psi = 2$, size is 0·068. There is a tendency for size to be subnominal for $J_1 > J_2$ and $\psi < 1$ or for $J_1 < J_2$ and $\psi > 1$. Romano's[38] asymptotic results support these conclusions. Regardless of the underlying distributions for $W_{ij\cdot}$, the permutation test will have the correct size asymptotically if either $J_1 = J_2$ or if $\text{var}(W_{ij\cdot}) = \text{var}(W_{2j\cdot})$, but not otherwise.

The unpaired t-tests and normal approximation test perform similarly to the permutation test (Table I). We rationalize the behaviour of the t-test with unbalanced designs as follows. Consider the case $J_1 = 6, J_2 = 3$ and $\psi = 5$. The pooled variance estimate of $\text{var}(U)$, namely $s^2\,(\frac{1}{J_1} + \frac{1}{J_2})$ has expectation $\text{var}(W_{1j\cdot})\ \{(J_1 - 1) + \psi(J_2 - 1)\}\ (J_1 + J_2 - 2)^{-1}\ (J_1^{-1} + J_2^{-1}) = 1{\cdot}071\ \text{var}(W_{1j\cdot})$. The true variance of U is $\text{var}(W_{1j\cdot})\,\{\frac{1}{J_1} + \frac{\psi}{J_2}\} = 1{\cdot}833\,\text{var}(W_{1j\cdot})$. Hence, the quantity t in equation (2) is larger than the $(1 - \alpha)$-quantile of the t distribution with probability that exceeds α. Miller[48] (Section 2.3) discusses these effects of unequal variance on the unpaired t-test, and Harwell *et al.*[49] summarize studies on the size of the F-test for analysis of variance under various violations of the model, including heteroscedasticity.

Unequal variances $\text{var}(W_{1j\cdot}) \neq \text{var}(W_{2j\cdot})$ can arise in several ways. Imagine a population of communities from which we have sampled the study population. For a sampled population assigned to treatment i, the mean response for an individual in community (i, j) is $E(W_{ijk}) = \mu_{ij}$ and the variance of individual responses within that community is σ_{ij}^2. Assuming we have chosen K_{ij} of the n_{ij} community members without replacement for a cohort study, the unconditional variance of $W_{ij\cdot}$ is

$$\text{var}(W_{ij\cdot}) = \text{var}(\mu_{ij}) + E(\sigma_{ij}^2/K_{ij})(n_{ij} - K_{ij})(n_{ij} - 1)^{-1}. \tag{4}$$

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Intervention $i = 1$ may be effective in some communities and deleterious in others, with the result that $\mathrm{var}(\mu_{1j})$ is large, and $\mathrm{var}(\mu_{1j}) > \mathrm{var}(\mu_{2j})$. This case corresponds to $\psi < 1$ in Table I. If there is more loss to follow-up in intervention than control communities, $(K_{1j} < K_{2j})$, then $\mathrm{var}(W_{1j\cdot})$ will tend to exceed $\mathrm{var}(W_{2j\cdot})$. Conversely, if $K_{1j} > K_{2j}$, $\mathrm{var}(W_{1j\cdot})$ will tend to be less than $\mathrm{var}(W_{2j\cdot})$. Because, however, the first component in equation (4) often exceeds the second component in community intervention trials, sample size disparities would need to be extreme to produce appreciable distortions of size for testing H_0.

These results suggest that, in the context of a community intervention trial in which the community level summary statistics $\{W_{ij\cdot}\}$ are approximately normally distributed, permutation tests likely yield nearly nominal level tests not only for testing the strong null hypothesis H_{0s}, but also for testing the weaker hypothesis of no average treatment effect, H_0, for balanced designs. This is good reason to use balanced designs. We discuss further the practical implications of these simulations in Section 6, but it is clear from Table I that size is near nominal levels unless there are both large imbalances in numbers of treated and control communities and appreciable differences in the variance.

3.2. Randomization-Based Inference: Pair-Matched Designs

In pair-matched designs, we pair community $(1,j)$ with community $(2,j)$ on the basis of demographic or other characteristics. Thus, j indexes a particular pair, and we use the paired differences $U_j = W_{1j\cdot} - W_{2j\cdot}$ to measure treatment effects. Permutational tests of the strong null hypothesis H_{0s} and test-based methods for calculating confidence intervals are available, as in Section 3.1. These paired tests are necessarily balanced and are more robust than unbalanced unpaired tests for the weak null hypothesis of no average treatment effect, H_0.

We generate the permutational distribution of $U = \frac{1}{J}\sum_{J=1}^{J} U_j$ by considering all 2^J possible ways of reassigning treatments within pairs. We need at least $J = 5$ pairs to have a critical region with significance level less than 0·05. A paired t-test on $J - 1$ degrees of freedom provides an excellent approximation[50] to the permutational distribution of U. The paired t statistic is

$$U\{s^2/J\}^{-1/2} \tag{5}$$

where $s^2 = \Sigma(U_j - U_\cdot)^2/(J-1)$.

Under the strong null hypothesis H_{0s}, each of these possible data rearrangements is equally likely. Now consider interventions that produce no change in average effect, so that $E(U_j) = 0$, but that alter other aspects of the distributions of $W_{1j\cdot}$ and $W_{2j\cdot}$. As long as the difference U_j remains symmetrically distributed about zero, the permutation test will have nominal size (except for discreteness).

For example, suppose the effect of treatment varies from community to community according to the model $W_{ijk} = \mu + \alpha_j + d_{ij} + \varepsilon_{ijk}$, where α_j is a fixed pair effect and d_{ij} is a random treatment effect with mean δ_i and variance σ_i^2. Then $U_j = W_{1j\cdot} - W_{2j\cdot} = d_{1j} - d_{2j} + \varepsilon_{1j\cdot} - \varepsilon_{2j\cdot}$ is symmetrically distributed about $\delta_1 - \delta_2$ and the U_j are independent provided:

1. $\{d_{ij}\}$ and $\{\varepsilon_{ijk}\}$ are independent;
2. d_{1j} and d_{2j} have the same distribution or are each symmetrically distributed; and
3. $\varepsilon_{1j\cdot}$ and $\varepsilon_{2j\cdot}$ have the same distribution or are each symmetrically distributed about a common location.

Under these circumstances, $P(U_1, U_2, \ldots, U_J) = P(\gamma_1 U_1, \gamma_2 U_2, \ldots, \gamma_J U_J)$ for any choices of $\gamma_j = \pm 1$, and we say the $\{U_j\}$ are 'orthant symmetric'.[50]

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In particular, if we suppose d_{1j} and d_{2j} are normally distributed but with different variances ϕ_1 and ϕ_2, then, for normal errors ε_{ijk}, U_j are symmetrically distributed about $\delta_1 - \delta_2$. Hence the paired permutation test yields a valid test of H_0: $\delta_1 = \delta_2$ as well as of the stronger null hypothesis, H_{0s}, that $d_{ij} = 0$ for all i and j. Thus the permutation test based on the paired design has nominal size for a wide class of treatment effect models, not only against the strong null hypothesis, H_{0s} that treatment has no effect on any community response $\{W_{ij\cdot}\}$, but also against the null hypothesis of no treatment effect on average. The permutation test for the unpaired design does not share this feature, as discussed in Section 3.1.

Even the paired permutation test, however, may not have nominal size under H_0 if intervention causes the distribution of U_j to become skewed. For example, suppose W_{ijk} is a Bernoulli variate and that in each pair of communities, j, the intervention community sample contains only $K_{1j} = 10$ subjects while the control community sample contains $K_{2j} = 100$ subjects. Suppose, further, that $E(W_{ijk}) = p = 0{\cdot}1$. Then $W_{1j\cdot}$ and $W_{2j\cdot}$ each has mean p, but $\mathrm{var}(W_{1j\cdot}) = p(1-p)/K_{1j} = 0{\cdot}009$, while $\mathrm{var}(W_{2j\cdot}) = 0{\cdot}0009$. Hence $W_{1j\cdot}$ and $W_{2j\cdot}$ have different distributions, both of which are skewed. The distribution of $W_{2j\cdot}$ is more nearly symmetric because $W_{2j\cdot}$ is based on a larger sample. Hence $U_j = W_{1j\cdot} - W_{2j\cdot}$ is skewed. The strong null hypothesis H_{0s} is not satisfied, because treatment is associated with the variance of $W_{ij\cdot}$. However, the weak null hypothesis that the average pairwise difference has expectation zero is satisfied. In terms of the previous model, we are assuming $\mu = \alpha_j = 0$, $d_{ij} = p$ is degenerate, and $\varepsilon_{ijk} = W_{ijk} - p$ has mean zero and variance $p(1-p)$. Note that ε_{ijk} is not symmetric unless $p = 0{\cdot}5$.

We present in Table II simulations of such Bernoulli data for several choices of K_{1j}, K_{2j} and J. These simulations correspond to extreme situations in which all intervention communities have larger sample sizes than control communities ($K_{1j} > K_{2j}$, $j = 1, 2, \ldots, J$) or all intervention communities have smaller sample sizes than control communities ($K_{1j} < K_{2j}$, $j = 1, 2, \ldots, J$). The nominal size of the permutation test for $J = 5$ is $2^{-5} = 0{\cdot}03125$, whereas for $J = 10$, 20 and 40, the nominal size is very near 0·050. Because the simulations have many repetitions, it is possible to demonstrate even small deviations from nominal size in Table II as being 'statistically significant'. We focus on situations in which the observed size exceeds the nominal size by 20 per cent or more. All tests are one-sided. Details for the simulation are in a footnote to Table II.

When $p = 0{\cdot}5$, the sizes of both the permutation test and paired t-test tend to be near nominal levels. We might have anticipated this result from the fact that the paired differences U_j are symmetrically distributed[50] with $p = 0{\cdot}5$.

For $p = 0{\cdot}1$, the paired t-test tends to have size that exceeds nominal levels when K_{1j} greatly exceeds K_{2j}. See, for example, the entries in Table II for $K_{1j} = 100$, and $K_{2j} = 9$ or 29 or the entries for $K_{1j} = 400$ and $K_{2j} = 39$. However, as J increases, the degree to which the size exceeds the nominal 0·05 level decreases. The central limit theorem implies that the paired t-test will have nominal size as $J \to \infty$ (see also Romano[38]). As K_{1j} and K_{2j} increase, even though the ratio of K_{1j} to K_{2j} remains large (for example, $K_{1j} = 100$, $K_{2j} = 9$ or $K_{1j} = 400$, $K_{2j} = 39$), the amount by which the paired t-test exceeds nominal size also diminishes. The paired one-sided t-test tends to have nominal or subnominal size when $K_{1j} < K_{2j}$.

One should distinguish these results for small J from those of Donner and Donald,[7] who found that the paired t-test has near nominal size for $K_{1j} = K_{2j}$, $j = 1, 2, \ldots, 6$ with $K_{11} = K_{12} = 20$, $K_{13} = K_{14} = 120$, and $K_{15} = K_{16} = 220$ and with $p = 0{\cdot}3$, 0·5 and 0·7. In the simulations of Donner and Donald, however, the sample sizes are the same in intervention and control communities, which implies that the differences U_j are symmetrically distributed. From Efron's[50] results on orthant symmetry, one can anticipate that the paired t-test has nearly nominal (or subnominal) size in such cases.

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Table II Estimated size (times 10,000) of one-sided permutation tests (upper numbers and paired t-tests (lower numbers) for Bernoulli probabilities $p = 0{\cdot}1$ and $p = 0{\cdot}5$†

		Number of pairs, J							
		5		10		20		40	
K_{1j}	K_{2j}	$p = 0{\cdot}1$	$p = 0{\cdot}5$	$p = 0{\cdot}1$	$p = 0{\cdot}5$	$p = 0{\cdot}1$	$p = 0{\cdot}5$	$p = 0{\cdot}1$	$p = 0{\cdot}5$
100	9	296	318	786*	494	698*	550	646*	484
		811*	525	817*	499	694*	552	636*	474
100	29	433*	323	621*	463	592	522	550	488
		638*	507	625*	471	602*	530	542	498
100	49	259	307	561	487	562	532	528	516
		516	495	549	488	562	520	530	520
100	99	226	323	483	547	594	540	452	494
		506	504	520	543	578	546	440	490
400	39	367	306	595	509	598	528	584	506
		642*	489	604*	510	602*	530	574	506
400	119	342	294	529	486	508	522	534	492
		523	514	543	492	526	530	530	492
9	100	293	296	303	506	316	518	400	510
		305	485	289	507	326	514	396	510
29	100	247	317	386	513	430	514	448	474
		388	519	387	521	434	506	452	476
49	100	345	332	430	468	486	494	512	454
		439	537	449	472	482	504	510	450
99	100	433*	323	522	488	460	478	506	490
		468	507	506	497	482	476	510	486
39	400	280	299	392	516	386	472	446	538
		419	502	380	526	384	468	444	540
119	440	275	324	452	534	466	560	480	436
		420	480	464	534	478	568	484	432
999	1000	347	320	NA	NA	NA	NA	NA	NA
		486	508						
9999	10000	324	311	NA	NA	NA	NA	NA	NA
		505	498						
Nominal size									
Permutation test	0·0312	0·0312	0·0498	0·0498	0·0500	0·0500	0·0500	0·0500	0·0500
Paired t-test	0·0500	0·0500	0·0500	0·0500	0·0500	0·0500	0·0500	0·0500	0·0500

* Asterisk indicates that observed size exceeds nominal level by 20 per cent or more

† Each entry is based on 10,000 repetitions except for $K_{1j} = 999$ or 9999, which are based on 100.000 repetitions and for $J = 20$ or 40, which are based on 5000 repetitions. Each pair of numbers corresponding to (K_{1j}, K_{2j}, J, p) represents an independent simulation study. NA means that the simulation was not performed. Calculations were performed in Fortran 77 (WATCOM complier) on a personal computer with a Pentioum™ microprocessor. Uniform pseudo-random numbers were supplied by the URAND subroutine in Fortran 77, based on the multiplicative-congruential method

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For $p = 0{\cdot}1$, the size of the paired permutation test, like the paired t-test, substantially exceeds nominal levels for $K_{1j} = 100$, $K_{2j} = 9$ and $J = 10, 20$ and 40. The size was also above nominal levels for $K_{1j} = 100$, $K_{2j} = 29$ and $J = 5$ and 10. Thus if sample sizes are consistently much larger in intervention than control communities and if $p < 0{\cdot}5$, the size of the permutation test may exceed nominal levels. In practice, however, these conditions are unlikely to arise because one would not ordinarily use very different cohort sizes in intervention and control communities. Moreover, if one assembles the cohorts before randomization, and if the extent of loss to follow-up is minimal or similar in intervention and control communities, the distributions of sizes of cohorts are similar in intervention and control communities.

With only $J = 5$ pairs, some surprising results appeared. Consider the entry $K_{1j} = 99$, $K_{2j} = 100$ and $J = 5$. The estimated size of the permutation test was 0·0433, well in excess of the nominal size, $(\frac{1}{2})^5 = 0{\cdot}03125$. We verified this result, and the results for all entries in Table II with $J = 5$ and K_{1j} and K_{2j} no larger than 100, as follows. The rejection region in this case is $\{(\hat{p}_{ij}): \hat{p}_{1j} > \hat{p}_{2j},\ j = 1, 2, 3, 4, 5\}$, where $\hat{p}_{ij}$ is the observed proportion of successes in intervention group i and pair j. The corresponding probability is $\{P(\hat{p}_{11} > \hat{p}_{12})\}^5$, because we assumed all pairs identical in this simulation. For $K_{1j} = 99$ and $K_{2j} = 100$, $P(\hat{p}_{11} > \hat{p}_{21}) = P(X_{11}/99 > X_{21}/100)$, where X_{11} and X_{12} are independent binomial variates with common success parameter $p = 0{\cdot}1$ and indices 99 and 100, respectively. This exact calculation yields actual size 0·045, which agrees well with the simulated result 0·0433. For $K_{1j} = 9999$ and $K_{2j} = 10{,}000$, the size is closer to the nominal level.

We defer the discussion of practical implications of these results until Section 6. However, the paired permutation test will have near nominal size under the weak null hypothesis in most circumstances that likely arise in practice. Only with small numbers of pairs ($J = 5$), a consistent pattern of different sample sizes in intervention and control communities, and an asymmetric distribution, such as the Bernoulli distribution with $p = 0{\cdot}1$, is there a realistic possibility that the size may substantially exceed nominal levels.

3.3. Adjustment for Covariates

A strength of permutational tests is that they are based on the randomization process and make no modelling assumptions, at least for testing H_{0s}. At the expense of modelling assumptions, however, one may use data on community level covariates Z and individual level covariates X to increase the power of permutational hypothesis tests and to adjust such tests for treatment imbalances on covariates. Examples of community level covariates are size of the community or average income in the community, whereas sex or age might represent individually measured covariates. We now review covariate adjustment procedures that have the potential to gain power if the covariate model is correct but that still yield valid tests of the null hypothesis, even if the posited model is incorrect. These ideas have been presented for clinical trials with simple randomization by Gail *et al.*[51] and used in the analysis of the COMMIT study.[10, 19]

3.3.1. Unpaired Designs

First, consider balanced unpaired designs. Let $h(\mu + Z_{ij}\beta_1 + X_{ijk}\beta_2)$ represent the hypothesized regression for an individual in community (i, j) with community-level covariates Z_{ij} and individual-level covariates X_{ijk} in the absence of treatment effects; here μ is an intercept parameter. It is understood that $h(.)$ is a known function and that Z_{ij} contains no treatment indicator. In the example below, we let $h(u) = \exp(u)\{1 + \exp(u)\}^{-1}$ describe the expectation of a Bernoulli variate as a logistic regression, but we handle other regressions similarly. It is possible to fit this model by regarding each individual as independent, leading to an estimator $\hat{W}_{ijk} = h(\hat{\mu} + Z_{ij}\hat{\beta}_1 + X_{ijk}\hat{\beta}_2)$.

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Under the 'very strong null hypothesis' H_{0s}^* that treatment has no effect on any W_{ijk} value, the residual $r_{ijk} = W_{ijk} - \hat{W}_{ijk}$ is independent of treatment, because $\hat{W}_{ijk}$ is a function only of covariates, which are independent of treatment by randomization, and of W_{ijk}, which is independent of treatment by hypothesis. Note that the hypothesis H_{0s}^* implies no treatment effect on any individual, whereas the previously mentioned H_{0s} implies no treatment effect on any community mean. Thus average residuals $r_{ij} = K_{ij}^{-1} \sum_{k=1}^{K_{ij}} r_{ijk}$ are exchangeable, provided $K_{ij} = K$, and we obtain an adjusted permutation test, U^{adj} from equation (1) by replacing $W_{ij\cdot}$ with $r_{ij\cdot}$. Even if the K_{ij} vary so that var($r_{1j\cdot}$) differs from var($r_{2j\cdot}$), the permutation test will have near nominal size in balanced designs (Table I). These results hold under H_{0s}^* even if we misspecify the regression function $h(.)$ or the terms in the model, and even if the covariates in the model are measured with error.

How much power we gain from such a procedure depends on the relative contributions from between community variation and variation attributable to measured covariates. As an example, consider the case where W_{ijk} is a Bernoulli outcome with expectation $h(\eta) = \exp(\eta)\{1 + \exp(\eta)\}^{-1}$, where

$$\eta_{ijk} = \mu + \delta T_i + X_{ijk}\beta_2 + V_{ij}. \tag{6}$$

In equation (6), $T_i = 1$ or 0 represents a treatment indicator and V_{ij} is a normally distributed random community effect, independent of T_i and X_{ijk}, with mean 0 and variance ϕ. The random effect V_{ij} represents unmeasured sources of variability among communities. We have assumed no important community level covariates Z in this example.

In the absence of intercommunity heterogeneity ($\phi = 0$), the hypothesized regression $h(\mu + \delta T_i + X_{ijk}\beta_2)$ is correct, and the efficient score for testing of $\delta = 0$ is half of $\sum_{jk} r_{1jk} - \sum_{jk} r_{2jk} = \sum_j K_{1j} r_{1j\cdot} - \sum_j K_{2j} r_{2j\cdot}$ If the sample sizes K_{1j} and K_{2j} are all equal and $J_1 = J_2$, we obtain the proposed statistic $U^{adj} = r_{1\cdot\cdot} - r_{2\cdot\cdot}$. Thus, U^{adj} provides an efficient test for $\delta = 0$ for balanced designs with equal cohort sizes in the absence of unmeasured intercommunity heterogeneity. As ϕ increases, covariate adjustment to control measured sources of intercommunity heterogeneity becomes relatively less important, and covariate adjustment yields smaller power gains.

To illustrate these ideas, we conducted simulations based on equation (6) with $X_{ijk} = 1$ if the subject was male and 0 if female (Table III).

Each row in Table III corresponds to an independent simulation study, each involving 3000 repetitions. We designed the twelve experiments on size to study the characteristics of the unadjusted and adjusted one-sided permutation tests for communities with varying sex compositions and sample sizes, as described in detail in the footnote to Table III. The estimated size of the tests based on U and U^{adj} were near the nominal 0·05 level in every experiment except experiment J, where the adjusted test was below the 95 per cent coverage interval (0·042, 0·058). When we repeated experiment J, the estimated size was 0·052. In these simulations, there is no systematic correlation between community size or covariate composition and treatment, because we assigned communities at random to treatment.

Studies on power (Table III) indicate that covariate adjustment has little effect if the covariate is not a predictor of outcome ($\beta_2 = 0$ in experiment M) or if all communities have the same covariate composition (experiment N). Adjustment, however, yields appreciable gains in power if covariate effects are strong and communities are heterogeneous in covariate composition (experiment O). Experiment Q demonstrates vitiation of the benefits of adjustment if there is a large uncontrolled component of between-community variation, represented by $\phi = 0·25$.

To relate these results on the usefulness of covariate adjustment to the concept of intraclass correlation, ρ, we use the generalized definition of intraclass correlation given by Commenges and

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Table III Simulations of size and power of unadjusted (U) and adjusted (U^{adj}) permutation tests based on unpaired designs with $J_1 = J_2 = 10$ community cohorts*

	Experiment	Model parameters					Estimated size or power‡	
		δ	ϕ	β_2	Proportion with $X = 1$ (male)†	Community sizes†	U	U^{adj}
Studies of size	A	0	0	0·693	(0·5, 0·5)	(300, 300)	0·055	0·055
	B	0	0·1	0·693	(0·5, 0·5)	(300, 300)	0·045	0·045
	C	0	0·25	0·693	(0·5, 0·5)	(300, 300)	0·049	0·049
	D	0	0	0·693	(0·5, 0·5)	(300, 100)	0·049	0·049
	E	0	0·1	0·693	(0·5, 0·5)	(300, 100)	0·053	0·053
	F	0	0·25	0·693	(0·5, 0·5)	(300, 100)	0·048	0·048
	G	0	0	0·693	(0·35, 0·65)	(300, 300)	0·042	0·048
	H	0	0·1	0·693	(0·35, 0·65)	(300, 300)	0·051	0·050
	I	0	0·25	0·693	(0·35, 0·65)	(300, 300)	0·054	0·053
	J	0	0	0·693	(0·35, 0·65)	(300, 100)	0·046	0·037
	K	0	0·1	0·693	(0·35, 0·65)	(300, 100)	0·048	0·045
	L	0	0·25	0·693	(0·35, 0·65)	(300, 100)	0·051	0·051
Studies of power	M	0·1	0	0	(0·35, 0·65)	(300, 300)	0·400	0·396
	N	0·1	0	1·099	(0·5, 0·5)	(300, 300)	0·466	0·466
	O	0·1	0	1·099	(0·35, 0·65)	(300, 300)	0·234	0·474
	P	0·1	0·1	1.099	(0·35, 0·65)	(300, 300)	0·143	0·151
	Q	0·1	0·25	1·099	(0·35, 0·65)	(300, 300)	0·111	0·111

* The parameter $\mu = -1{\cdot}7346$ in equation (6). Other parameter defined in the table. We used GAUSS (version 3.2) on a personal computer with a Pentium™ microprocessor to perform these calculations. Uniform and normal pseudo-random numbers were obtained by a multiplicative-congruential method (procedure RNDUS) and by an acceptance-rejection method (RNDNS), respectively.

† The notation (0·5, 0·5) indicates that all cohorts had 50 per cent males, wheras the notation (0·35, 0·65) indicates that 10 cohorts had 35 per cent males and 10 cohorts had 65 per cent males. Likewise the notation (300, 300) means that 20 cohorts had about 300 subjects, whereas (300, 100) means that 10 cohorts had about 300 members and 10 cohorts had about 100 members. The combined notation (0.35, 0.65) and (300, 100) means that 5 cohorts were 35 per cent male and had about 300 members, 5 were 35 per cent male with about 100 members, 5 were 65 per cent male with about 300 members and 5 were 65 per cent male with about 100 members. Such community cohorts were allocated to treatment at random from among $20!/(10!)^2$ possible assignments. To avoid ties, the actual sizes of cohorts with about 300 members were 281, 283, 285, ... 319 for (300, 300) and were 281, 287, 289, 295, 297, 303, 305, 311, 313, 319 for (300, 100). The actual numbers with about 100 for (300, 100) were 83, 85, 91, 93, 99, 101, 107, 109, 115, 117

‡ One-sided 0·05 level tests were used. Estimates are based on 3000 repetitions

Jacqmin,[52] namely $\rho = \mathrm{var}\{E(W_{ijk}|V_{ij})\}/\mathrm{var}(W_{ijk})$. For fixed T_i and X_{ijk}, Taylor's series expansion yields the approximate result $\rho = (pq)^2\phi\{(pq)^2\phi + pq\}^{-1}$, where $p = 1 - q = h(\mu + \delta T_i + X_{ijk}\beta)$. For control communities ($T_i = 0$) with X_{ijk} set at its average value, 0·5, we note that $X_{ijk}\beta = 0{\cdot}5495$, and $\mu = -1{\cdot}7346$. One obtains $p = 0{\cdot}23414$ and $\rho = 0{\cdot}018$ and 0·043, respectively, for $\phi = 0{\cdot}1$ and $\phi = 0{\cdot}25$. Thus the values $\phi = 0{\cdot}1$ and 0·25 in experiments P and Q (Table III) correspond to substantial, but not unrealistic,[53] degrees of intraclass correlation.

To summarize, these simulations indicate that the size of the adjusted permutation test U^{adj} is near nominal levels under the very strong null hypothesis H^*_{0s}. Gains in power from adjustment are modest, especially if there is appreciable intercommunity heterogeneity not represented by measured covariates.

In principle, one can adjust on community-level covariates by adding $Z_{ij}\beta_1$ to equation (6). Note, however that $\{W_{ij\cdot}\}$ are contained in a vector space $\mathscr{V}(IJ)$ of dimension IJ. The unadjusted residuals $(W_{ij\cdot} - W_{\cdots})$ used to compute $U = \Sigma(W_{1j\cdot} - W_{\cdots}) - \Sigma(W_{2j\cdot} - W_{\cdots}) = \sum_j W_{1j\cdot} - \sum_j W_{2j\cdot}$ for balanced designs are contained in $\mathscr{V}(IJ - 1)$. Suppose that $K_{ij} = K$.

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Score equations for β_1,

$$\sum Z_{ij}\,[W_{ijk} - \exp(\hat{\mu} + Z_{ij}\hat{\beta}_1 + X_{ijk}\hat{\beta}_2)\,\{1 + \exp(\hat{\mu} + Z_{ij}\hat{\beta}_1 + X_{ijk}\hat{\beta}_2)\}^{-1}] = 0$$

impose additional constraints on the quantities $r_{ij\cdot}$, so that $r_{ij\cdot}$ are contained in a space of dimension $(IJ - 1 - \dim)$, where dim is the dimension of $\{Z_{ij}\}$. If dim equals or exceeds $IJ - 1$, the permutational distribution of $\{r_{ij\cdot}\}$ is degenerate. These arguments suggest that one should adjust at most for a few very important community-level covariates.

3.3.2. *Paired Designs*

To obtain a covariate adjustment for paired designs, we used the previous Bernoulli model but with expectation

$$h(\mu_j + \delta\,T_i + \delta\,T_i + Z_{ij}\beta_1 + X_{ijk}\beta_2) \tag{7}$$

where parameters μ_j correspond to pairs $j = 1, 2, \ldots, J$. There are J constraints, $r_{1j\cdot}K_{1j} + r_{2j\cdot}K_{2j} = 0$, that represent score equations for μ_j. The resulting score for testing $\delta = 0$ is $\sum_j r_{1j\cdot}K_{1j} = 0{\cdot}5 \times \Sigma(r_{1j\cdot}K_{1j} - r_{2j\cdot}K_{2j})$. If $K_{ij} = K$, this score equation is equivalent to $\sum_j(r_{1j\cdot} - r_{2j\cdot})$, and we obtain the adjusted permutation test by replacing $W_{1j\cdot} - W_{2j\cdot}$ with $r_{1j\cdot} - r_{2j\cdot}$ in the definition of U_j in Section 3.2. Note that $r_{1j\cdot} - r_{2j\cdot}$ is a valid basis for a permutation test even if K_{ij} varies.

Under the strong null hypothesis H_{0s}^* that treatment has no effect on any individual W_{ijk} value, treatment is independent of r_{ijk} and hence of $r_{1j\cdot} - r_{2j\cdot}$, as argued previously. Hence, if $K_{1j} = K_{2j}$, treatment is exchangeable within pairs, and the permutation test is justified. Unless sample sizes K_{ij} are associated with treatment, as described in Section 3.2 and Table II, one can anticipate that the size of the permutation test will have near nominal levels.

To examine the size and power of the adjusted test, we conducted simulations (Table IV). As for unpaired tests, we used the model (6), except we replaced μ with μ_j in community pair $j, j = 1, 2, \ldots, 10$. In these experiments, V_{ij} corresponds to random heterogeneity of communities within pairs. Each row in Table IV corresponds to an independent simulation study that involved 10,000 repetitions. We designed the six experiments on size to examine the performance of the unadjusted and adjusted one-sided 0·05 (actually 0·0498) level permutation tests for cohort pairs with varying numbers of subjects, various covariate compositions and various levels of between-community within-pair random heterogeneity (ϕ). Both the unadjusted (U) and adjusted (U^{adj}) permutation tests had size very near nominal levels (Table IV, rows A to F).

Experiments G, H, I, J and K (Table IV) describe power gains from adjustment for an important covariate. There is no gain in power if the covariate has no effect on response (experiment G). Likewise, if the covariate distribution is the same in both cohorts in a community pair, there is no gain in power from adjustment (experiment H). Covariate adjustment increases power from 0·166 to 0·428 if there is no background random heterogeneity ($\phi = 0$) between communities in a pair (experiment I). If there is a large component of random between-community variation ($\phi = 0{\cdot}25$ corresponds to within-pair intraclass correlation 0·043), the power of the unadjusted test is 0·095 and the power of the adjusted test is only 0·099 (experiment K). Thus, in the presence of substantial random community heterogeneity, there is little power gained by adjusting for known covariates.

Although it is possible to adjust for community level covariates Z_{ij} in equation (7), the quantities $r_{1j\cdot} - r_{2j\cdot}$ are constrained to a space of dimension J even before such adjustment. Therefore, one must take care not to include too many such covariates.

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Table IV. Simulations of size and power of unadjusted (U) and adjusted (U^{adj}) permutation tests based on paired designs with $J_1 = 10$ pairs of communities*

Experiment		Model parameters					Estimated size or power‡	
		δ	ϕ	β_2	Proportion with $X = 1$ (male)†	Cohort sizes†	U	U^{adj}
Studies of size	A	0	0	0·693	(0·5, 0·5)	(300, 300)	0·051	0·052
	B	0	0·25	0·693	(0·5, 0·5)	(300, 300)	0·054	0·054
	C	0	0	0·693	(0·35, 0·65)	(300, 300)	0·049	0·049
	D	0	0·25	0·693	(0·35, 0·65)	(300, 300)	0·050	0·051
	E	0	0	0·693	(0·5, 0·5)	(300, 100)	0·049	0·050
	F	0	0·25	0·693	(0·5, 0·5)	(300, 100)	0·050	0·049
Studies of power	G	0·1	0	0·000	(0·35, ·0·65)	(300, 300)	0·343	0·341
	H	0·1	0	1·099	(0·5, 0·5)	(300, 300)	0·426	0·426
	I	0·1	0	1·099	(0·35, 0·65)	(300, 300)	0·166	0·428
	J	0·1	0·1	1·099	(0·35, 0·65)	(300, 300)	0·129	0·153
	K	0·1	0·25	1·099	(0·35, 0·65)	(300, 300)	0·095	0·099

* The parameters μ_j for pair j were $-2{\cdot}94444$, $-2{\cdot}58669$, $-2{\cdot}31363$, $-2{\cdot}09074$, $-1{\cdot}81529$, $-1{\cdot}65823$, $-1{\cdot}51635$, $-1{\cdot}38629$, $-1{\cdot}26567$ and $-1{\cdot}09861$. Other parameters are defined in the table. Calculations and random numbers were obtained as for Table III

† The notation is as in footnote of Table III except that it refers to pairs of communities. For example (0·35, 0·65), (300, 300) means that one community of each pair has 35 per cent males ($X = 1$) and the other community has 65 per cent males, and the total cohort size is about 300 in each community cohort in the pair. Within each pair, communities were allocated to treatment at random. To avoid ties, the actual sizes of cohorts with about 300 subjects were 281, 287, 289, 295, 297, 303, 305, 311, 313, 319, and for cohorts with about 100 subjects, the actual sizes were 83, 85, 91, 93, 99, 101, 107, 109, 115, 117

‡ A one-sided 0·05 (actually 0·0498) level test was used. Estimates were based on 10,000 repetitions

4. MISSING DATA IN COHORT STUDIES

Loss to follow-up of cohort members can threaten the validity of an otherwise sound study. If those lost to follow-up have the same distribution of responses, W_{ijk}, as those who remain on study, we say the lost data are missing completely at random (MCAR).[44] In this case, an analysis of only those subjects with complete data yields valid results. From Sections 3.1 and 3.2, balanced unpaired and paired randomization tests will have near nominal size unless there is a large disparity in extent of loss to follow-up between intervention and control communities. This is unlikely to occur, because the extent of loss to follow-up is often determined by factors unrelated to intervention, such as socioeconomic status of cohort members, nature of the informed consent procedure, completeness of initial identifying information, plans to maintain contact with cohort members during the study, and methods used to find those with whom one has lost contact. Therefore, if one can accept the MCAR assumption, one can confidently apply randomization tests to those with complete data.

A serious concern is that those lost to follow-up are not representative of their respective cohorts. Suppose in a smoking cessation trial that those lost to follow-up in the intervention communities are over-represented among non-quitters, whereas those lost to follow-up in the control communities are representative of control cohort members. In this case, the observed quit rate among those with complete data in intervention communities overestimates the true quit rate in intervention cohorts, whereas the observed quit rates among subjects with complete data in control communities is unbiased for the true quit rates in control communities. Thus the

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estimated difference in quit rates between intervention and control communities is upwardly biased. This is an example of 'differential non-response bias', because non-response bias is present in intervention but not in control communities. Randomization tests provide no protection against differential non-response bias.

It is possible that the bias from loss to follow-up in the treated communities is exactly balanced by a comparable bias from loss to follow-up in control communities. In this case, estimates of treatment effect are unbiased, and we say there is 'non-differential non-response bias'. Under the assumptions of no treatment effect and non-differential non-response bias, randomization tests have near nominal size.

Sometimes one has intercurrent response data that one can use to obtain valid randomization tests under weaker assumptions than the MCAR assumption. For example, suppose that we have measured smoking status at years 2 and 5 of follow-up and that the main endpoint of the study, W_{ijk}, is whether or not the subject has quit at year 5. Among those with complete data at years 2 and 5, we can estimate $\hat{p}$, the overall quit rate at year 5, $\hat{p}_q$, the quit rate at year 5 among those who had quit at year 2, and $\hat{p}_s$, the quit rate at year 5 among those still smoking at year 2. We are reluctant to make the assumption that subjects still smoking at year 2 but who have subsequently become lost have the overall quit rate $\hat{p}$. Instead, we might assume that such subjects had the same quit rate, $\hat{p}_s$, as others still smoking at year 2. Likewise, we assume that individuals who had quit at year 2 and who have subsequently become lost have a final quit rate $\hat{p}_q$, and we assume that those lost before year 2 have a quit rate $\hat{p}$. One can show that these assumptions, which we call MAR*, lead to unbiased estimates of cohort quit rates. Moreover, the MAR* assumptions are true whenever the MCAR assumption is valid. (The MAR* assumptions are slightly stronger than the missing at random assumption.[44, 54]) Procedures of this type using baseline covariates as well as intercurrent response data were employed to impute cohort quit rates for the COMMIT study,[19] and unreported simulations confirmed the validity of randomization tests based on the pairwise difference of imputed cohort quit rates under these assumptions.

One should make every effort to minimize loss to follow-up in community intervention trials, in which the tendency to lose contact with subjects is often greater than in conventional clinical trials. One factor that contributes to the tendency to lose contact with subjects in community intervention trials is the necessity to treat cohort members no differently from other community members. Thus, cohort members may not have the same sense of loyalty to the trial as in conventional, individually-randomized clinical trials.

5. PERMUTATIONAL ANALYSIS OF CROSS-SECTIONAL STUDIES

Because cross-sectional studies yield data from independent samples at baseline (t_0) and subsequent times, we must base summary measures of outcome for a given community on the marginal distributions of responses at these times. For simplicity, assume measurements are taken only at times t_0 and t_1. Then the summary measure for community j on treatment i might be $W^*_{ij\cdot} = Y^*_{ij\cdot t_1} - Y^*_{ij\cdot t_0}$ or $W^*_{ij\cdot} = \log\{Y^*_{ij\cdot t_1}/(1 - Y^*_{ij\cdot t_1})\} - \log\{Y^*_{ij\cdot t_0}/(1 - Y^*_{ij\cdot t_0})\}$. Under the strong null hypothesis that treatment has no effect on any $W^*_{ij\cdot}$, one can perform randomization tests on $\{W^*_{ij\cdot}\}$ as in Section 2. The results in Section 2 also indicate that the size of randomization tests based on $\{W^*_{ij\cdot}\}$ are near nominal levels in most cases under the weaker null hypothesis that treatment has no effect on the expected value of $W^*_{ij\cdot}$.

It may be desirable to adjust $W^*_{ij\cdot}$ to eliminate the effects of temporal changes in age and sex composition. For example, in the COMMIT study,[40] crude estimates of initial (1988) and final (1993) smoking prevalence, $Y^*_{ij\cdot t_0}$ and $Y^*_{ij\cdot t_1}$, were replaced by direct standardized prevalence rates. Census data on the 1990 age and sex distribution for community j on intervention i were used to

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standardize the initial and final prevalence estimates. After standardization, $W^*_{ij.}$ (adjusted) = adjusted final prevalence – adjusted initial prevalence, represented the change in smoking prevalence not due to temporal changes in age or sex composition. One can perform permutation tests on the age and sex adjusted quantities $W^*_{ij.}$ (adjusted) to test the strong null hypothesis that treatment has no effect on these quantities.

If a substantial number of sampled individuals refuse to answer survey questions, cross-sectional studies can lead to biased estimates of treatment effect. Such non-response bias in the estimate of treatment effect may be minimal if the extent of bias is similar in intervention and control communities. Standardization by age and sex can reduce bias if respondents within a given sex and age level are representative of that sex and age category. In this case, the standardized prevalence estimate based on responders is unbiased, even though the crude prevalence rate may suffer non-response bias.

6. DISCUSSION

In this paper we have stressed the importance of scientific objectives in deciding whether to adopt a longitudinal or cross-sectional design, the former being more informative when interest focuses on assessing the effects of intervention on individuals (Section 2). We have also considered other aspects of design and analysis, including whether to match communities, the difficulty of obtaining information needed to design such studies, the importance of endpoint selection (Section 2), and loss to follow-up or non-response to surveys (Sections 2.3, 4 and 5). Many of these comments reflect experiences in the design and analysis of the recently published COMMIT study to promote smoking cessation. That study included both an assessment of smoking quit rates in cohorts[19] and an assessment of changes in smoking prevalence based on initial and final cross-sectional surveys.[40]

A second theme of this paper is the study of the properties of randomization tests, including randomization tests with covariate adjustment. These procedures were used in the pair-matched COMMIT study,[19] but we have examined their statistical properties both for pair-matched (Sections 3.2 and 3.3) and unmatched (Sections 3.1 and 3.3) designs.

An important issue is whether randomization tests have proper size under the weak null hypothesis that treatment produces no effect on average[32] and under conditions likely to arise in community intervention trials. For example, the mean community responses among intervention and control communities might be the same, but the variances of the responses might differ. For unmatched designs, the size is near nominal levels, provided the numbers of intervention and control communities are equal (balanced design), whether or not the variances of community responses differ among intervention or control communities (Table I). If there are fewer intervention than control communities, however, size can exceed nominal levels if the variance of community responses is larger among intervention than among control communities (Table I). If there are more intervention than control communities, size can exceed nominal levels if the variance among intervention community responses is smaller than among control community responses (Table I). We found analogous results for the unpaired t-test.[48] Although it may be uncommon that differences in variances are sufficiently severe to produce appreciable distortion in the size of randomization tests, we recommend use of equal numbers of intervention and control communities to avoid this potential problem.

Pair-matched designs are necessarily balanced, and the size of the corresponding randomization tests are ordinarily near nominal levels under the weak null hypothesis. It is sufficient that the pairwise differences in community responses are independently and symmetrically distributed or that there are many independent community pairs. We have identified only one circumstance

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in which the size of the pair-matched randomization test can exceed nominal levels appreciably under the weak null hypothesis. If there is only a small number of pairs (that is, $J = 5$), and if intervention cohorts are consistently larger or consistently smaller than control cohorts, and if the individual response distribution is skewed (such as a Bernoulli response with probability $p = 0{\cdot}1$), then size can exceed nominal levels (Table II). Proper design can avoid, or make unlikely, the first and second of these conditions.

Thus, in most practical settings, the size of randomization tests are near nominal levels, not only under the strong null hypothesis of no treatment effect on any community, but also under the weak null hypothesis of no treatment effect on average.

We have also studied a procedure for covariate adjustment for use with randomization tests of no treatment effect (Section 3.3). This procedure is based on the assumption that treatment has no effect on individuals' responses. This procedure was used to analyse the pair-matched COMMIT cohort study.[10, 19] Simulations for unmatched (Table III) and matched (Table IV) designs indicate that such procedures have proper size. This is true whether or not the regression model used for covariate adjustment is correct. One can achieve gains in power from covariate adjustment when covariate effects are strong and when there is substantial heterogeneity of covariate distributions among communities (Tables III and IV). Such gains in power from covariate adjustment, however, become attenuated when there is appreciable random community heterogeneity (Tables III and IV) that induces residual intraclass correlation of individual responses within communities.

One needs additional assumptions to justify randomization tests (or any procedure) when response data are missing (Sections 4 and 5).

Statistical models that make additional distributional assumptions can provide an alternative basis for hypothesis testing in cohort studies. For example, an analysis of variance model with random community effects and fixed treatment effect leads[8, 9] to the unpaired t-test in Section 3.1. Another normal model with random community effects but a separate pair-specific fixed intercept parameter for each community pair and with fixed treatment effect leads to the paired t-test in Section 3.2. Such mixed models have also motivated analyses of cross-sectional designs.[8, 9, 11] Extension of such mixed effects models to include individual and community-level covariate effects have been proposed.[15] 'Marginal' regression models do not require a random community effect but allow for intra-community correlations by estimating required variances empirically.[17, 29, 31] Recent simulation studies indicate, however, that the size of such procedures may exceed nominal levels unless there are a large number of communities.[55, 56] Further work is needed to explore the strengths and weaknesses of such model-based approaches, compared to randomization-based procedures.

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